



Cambridge (CIE) IGCSE Maths: Extended



Your notes

Right-Angled Triangles (Pythagoras & Trigonometry)

Contents

- * Pythagoras Theorem
- * SOHCAHTOA
- * Angles of Elevation & Depression
- * Exact Trig Values



Your notes

Pythagoras Theorem

Pythagoras Theorem

Who is Pythagoras?

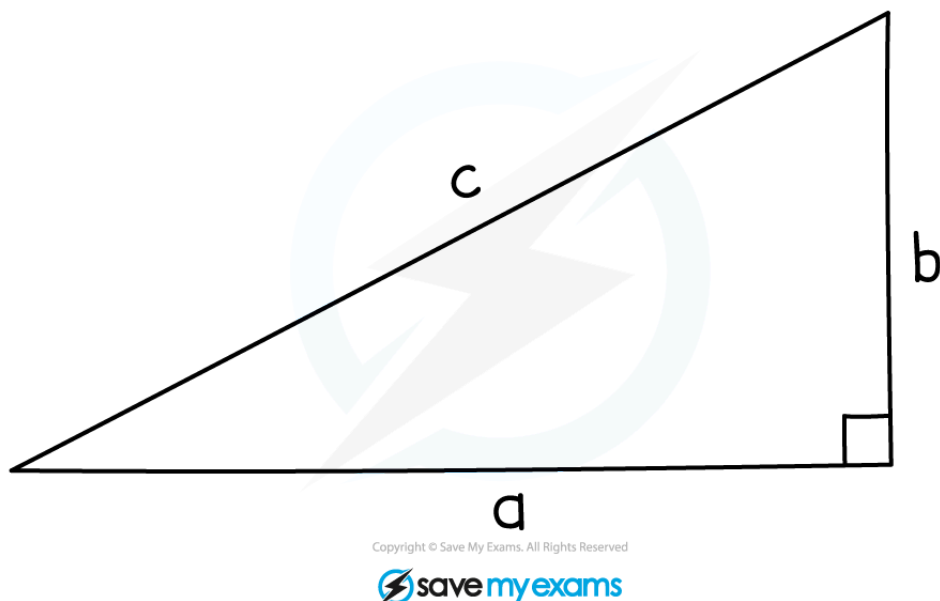
- **Pythagoras** was a **Greek mathematician** who lived over 2500 years ago
- He is most famous for **Pythagoras' theorem**, which includes the important formula for **right-angled triangles**

What is Pythagoras' theorem?

- **Pythagoras' theorem** is a formula that links the **lengths** of the **three sides** of a **right-angled triangle**
- The **longest side** of a right-angled triangle is called the **hypotenuse**
 - The hypotenuse will always be the side **opposite** the right angle
- **Pythagoras' theorem** states that $a^2 + b^2 = c^2$
 - **c** is the **length** of the **hypotenuse**
 - **a** and **b** are the **lengths** of the **two shorter sides**
 - It does **not matter** which is labelled **a** and which is labelled **b**



Your notes



How do I use Pythagoras' theorem to find the length of the hypotenuse?

- To find the length of the **hypotenuse**
 - **Square** the lengths of the **two shorter sides**
 - **Add** these two numbers together
 - Take the positive **square root**
- This can be written as $c = \sqrt{a^2 + b^2}$
 - This is just a **rearrangement** of the formula $a^2 + b^2 = c^2$ to make **c** the subject
 - Note that when finding the hypotenuse you **add** inside the square root

How do I use Pythagoras' theorem to find the length of a shorter side?

- To find the length of a **shorter side**
 - **Square** the lengths of the **hypotenuse** and the **other shorter side**
 - **Subtract** these numbers to find the difference

- Take the positive **square root**
- This can be written as $a = \sqrt{c^2 - b^2}$
 - This is just a **rearrangement** of the formula $a^2 + b^2 = c^2$ to make a the subject
 - Note that when finding one of the shorter sides you **subtract** inside the square root



Examiner Tips and Tricks

- If the hypotenuse ends up being shorter than another side in your answer then you have made a **mistake** somewhere
- Make sure that you subtract the **smaller value from the bigger value** when finding the length of a shorter side
 - Otherwise you will get a "**Math Error**" when trying to find the square root of a negative number
- In questions with **multiple steps**:
 - Leave your answer as an **exact answer**
 - **Do not round** until the very **end of the question**



Worked Example

In the following diagram:

$$AB = 12 \text{ cm}$$

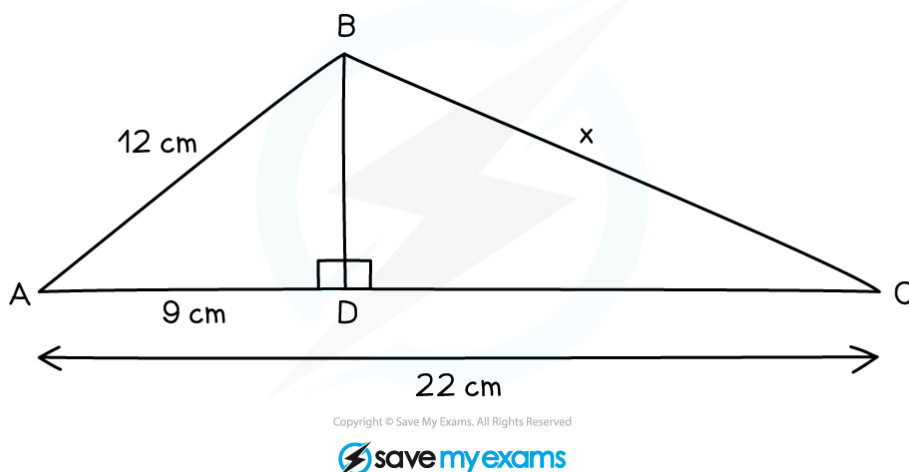
AC is a straight line, with $AD = 9 \text{ cm}$ and $AC = 22 \text{ cm}$



Your notes



Your notes



Find x , the length of side BC . Give your answer to 1 decimal place.

To find x , we first need to find the length of BD using triangle ABD

Note that BD is a shorter side

Apply Pythagoras' theorem, $a = \sqrt{c^2 - b^2}$

$$BD = \sqrt{12^2 - 9^2} = \sqrt{63} = 7.93725\dots$$

It is best to leave rounding until the very end, use $\sqrt{63}$ (or $3\sqrt{7}$ if this is what your calculator has given you) in subsequent working

Find the length of DC by subtracting the length of AD from the length of AC

$$DC = 22 - 9 = 13 \text{ cm}$$

Now we can find x using triangle BCD

Note that BC is the hypotenuse

Apply Pythagoras' theorem, $c = \sqrt{a^2 + b^2}$

$$x = \sqrt{BD^2 + DC^2} = \sqrt{(\sqrt{63})^2 + 13^2} = \sqrt{63 + 169}$$

$$x = \sqrt{232} = 15.23154621\dots$$

Round to 1 decimal place

$$x = 15.2 \text{ cm}$$



Your notes



Your notes

SOHCAHTOA

SOHCAHTOA

What is trigonometry?

- Trigonometry is the mathematics of **angles** in triangles
- It looks at the relationship between **side lengths** and **angles of triangles**

What are sin, cos and tan?

- The **three** trigonometric functions **sine**, **cosine** and **tangent**
 - They come from **ratios of side lengths** in **right-angled** triangles
- You must **label the sides** of a **right-angled triangle** in relation to a **chosen angle θ**
 - The **hypotenuse**, **H**, is the **longest side** in a right-angled triangle
 - It will always be **opposite** the right angle
 - The side **opposite θ** will be labelled **opposite**, **O**
 - The side **next to θ** will be labelled **adjacent**, **A**
- The functions sine, cosine and tangent are the ratios of the lengths of these sides as follows

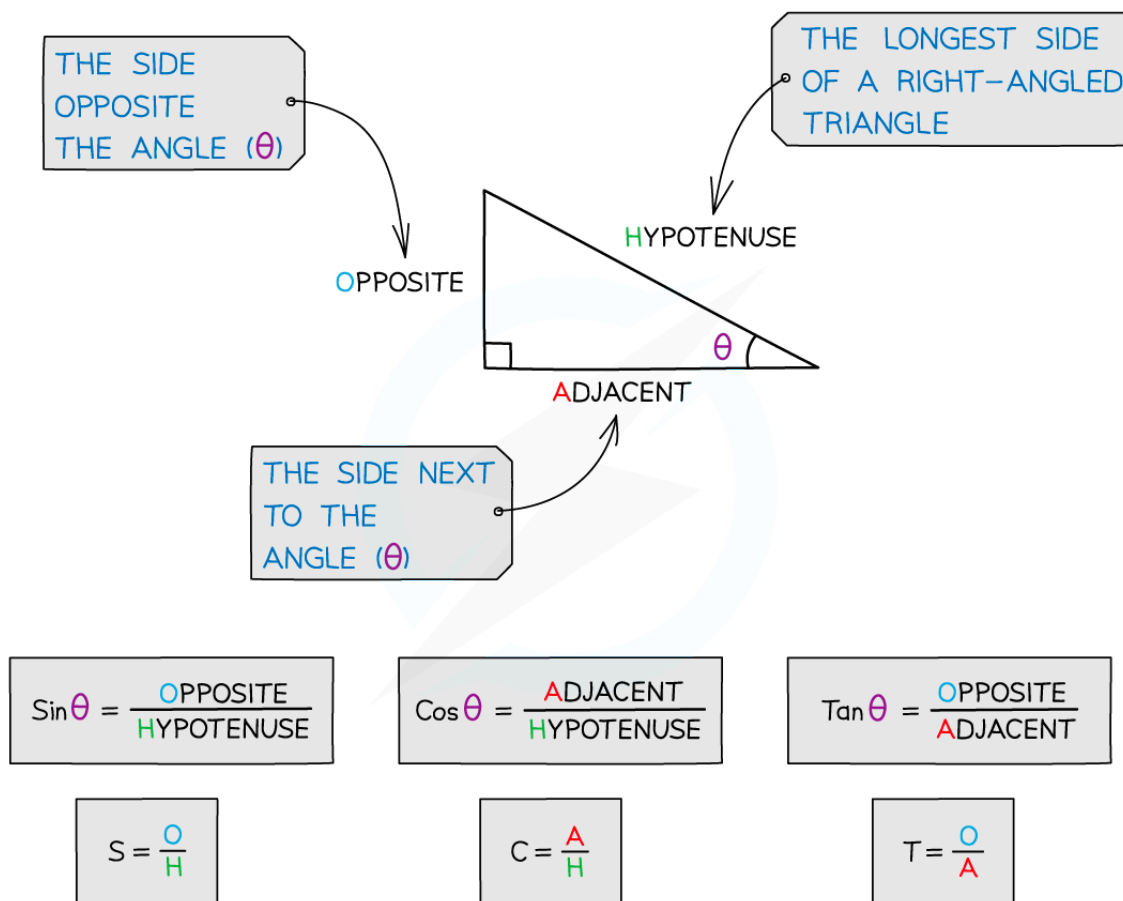
$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{O}{H}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{A}{H}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{O}{A}$$



Your notes



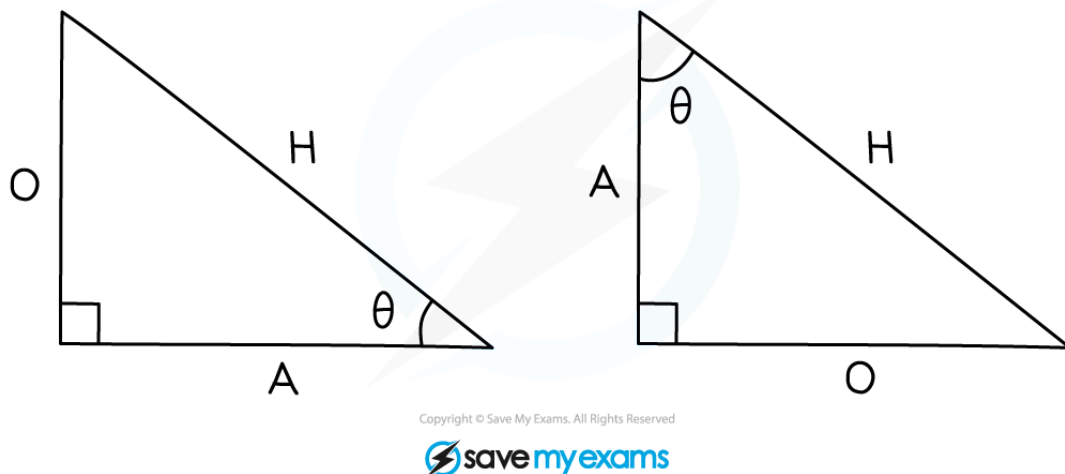
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What is SOHCAHTOA?

- SOHCAHTOA is a mnemonic often used to remember which ratio is which
 - Sin is Opposite over Hypotenuse
 - Cos is Adjacent over Hypotenuse
 - Tan is Opposite over Adjacent
- H is always the same but O and A change depending on which angle is labelled as θ



Your notes



How can I use SOHCAHTOA to find missing lengths?

STEP 1

Label the sides of the triangle as H, O and A

STEP 2

Identify which **trigonometric ratio** to use: sin, cos or tan

- Write down the letter of the **length** you are **given**
- Write down the letter of the **length** you **want to find**
- Find the two letters in **SOHCAHTOA** to identify which ratio to use
 - If you have **A** and **H** then use **cos**

STEP 3

Substitute the values into the relevant trigonometric formula

- Remember to put brackets around the angle

$$\sin(50) = \frac{A}{7} \text{ or } \cos(40) = \frac{3}{H}$$

STEP 4

Rearrange and **solve** for the **unknown letter**

- You will either need to **multiply** or **divide**



Your notes

- $\sin(50) = \frac{A}{7}$ leads to $A = 7 \times \sin(50)$

- $\cos(40) = \frac{3}{H}$ leads to $H = \frac{3}{\cos(40)}$

- **STEP 5**

Type the expression into your calculator

- The **question** might ask you to **round your answer**
- If not then round to **three significant figures**

How can I use SOHCAHTOA to find missing angles?

- **STEP 1**

Label the sides of the triangle as H, O and A

- **STEP 2**

Identify which **trigonometric ratio** to use: sin, cos or tan

- Write down the letters of the **lengths** you are **given**
- Find the two letters in **SOHCAHTOA** to identify which ratio to use
 - If you have **O** and **A** then use **tan**

- **STEP 3**

Substitute the values into the relevant trigonometric formula

- The **angle** will be **unknown**

- $\tan(\theta) = \frac{3}{4}$

- **STEP 4**

Substitute the fraction into the **inverse trigonometric function**

- You normally need to press **SHIFT** on your calculator first

- $\tan(\theta) = \frac{3}{4}$ leads to $\theta = \tan^{-1}\left(\frac{3}{4}\right)$

- **STEP 5**

Type the expression into your calculator

- The **question** might ask you to **round your answer**
- If not then round to **one decimal place**



Your notes

How do I find the shortest distance from a point to a line?

- The shortest distance from any point to a line will always be the **perpendicular** distance
- Form a right-angled triangle and then use SOHCAHTOA to find the relevant distance



Examiner Tips and Tricks

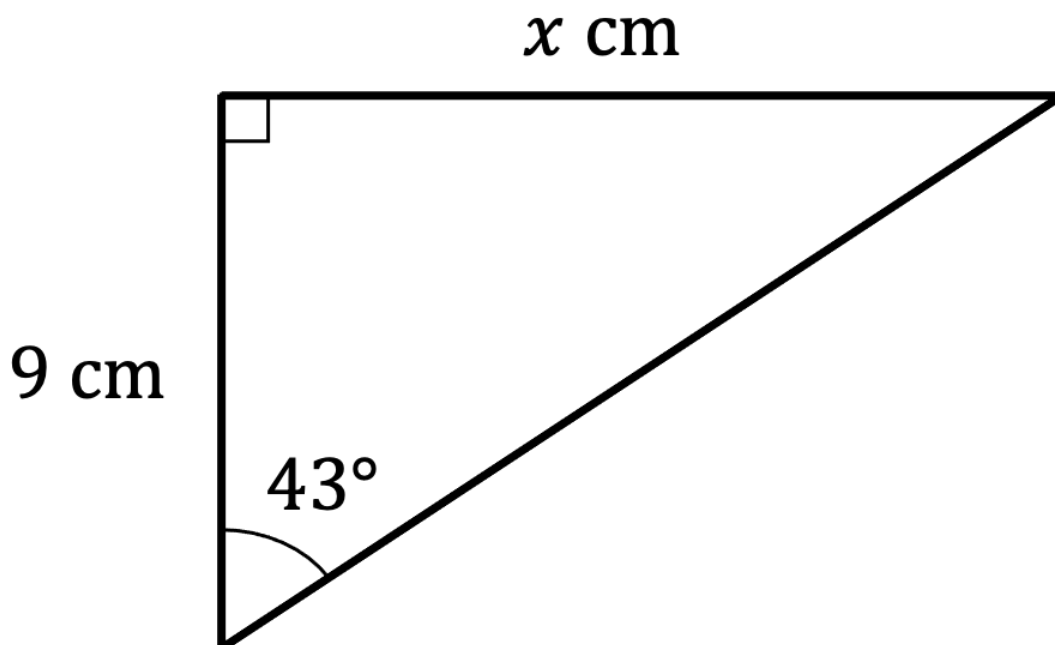
- SOHCAHTOA (like Pythagoras) can only be used in **right-angled triangles**
- Ensure your calculator is set to measure angles in **degrees**
 - You should see the letter **D** or the word **Deg** at the top of your screen



Worked Example

Find the length of the side x cm in the following triangle.

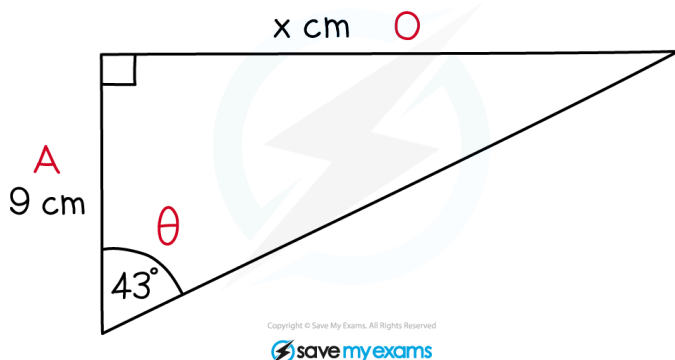
Give your answer to 3 significant figures.





Your notes

First label the triangle



We know A and we want to know O - that's TOA or $\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$

$$\tan(43) = \frac{x}{9}$$

Multiply both sides by 9

$$9 \times \tan(43) = x$$

Enter on your calculator

$$x = 8.3926...$$

Round to 3 significant figures

$$x = 8.39 \text{ cm}$$



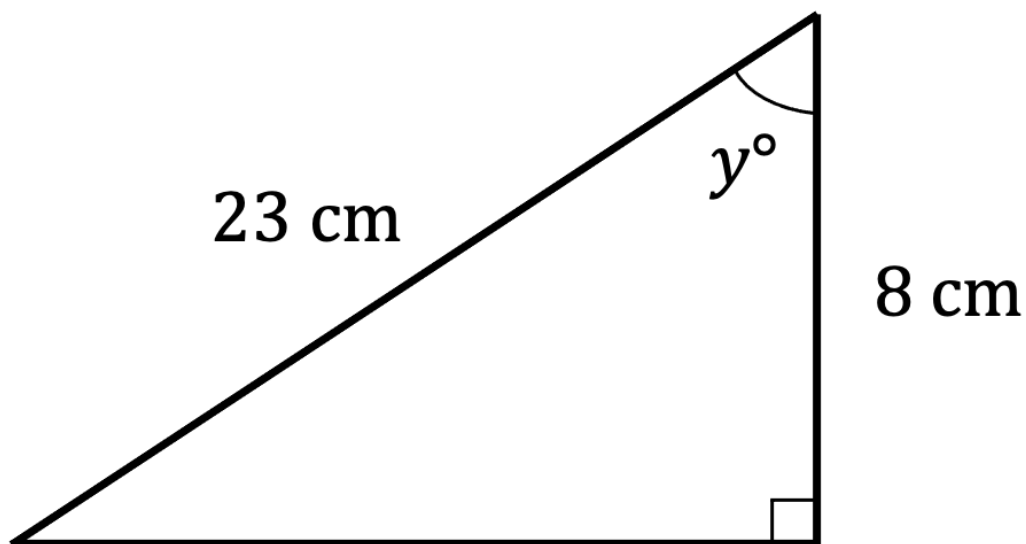
Worked Example

Find the value of the angle y° in the following triangle.

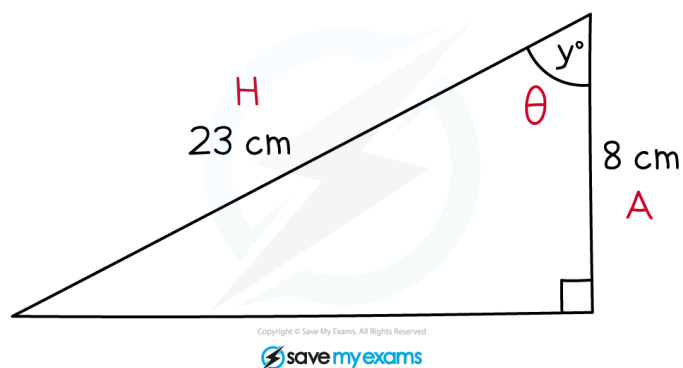
Give your answer to 1 decimal place.



Your notes



First label the triangle



We know A and H - that's CAH or $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$

$$\cos(y) = \frac{8}{23}$$

Use inverse cos to find y

$$y = \cos^{-1}\left(\frac{8}{23}\right)$$

Enter on your calculator

$$y = 69.6455\dots$$

Round to 1 decimal place

$$y = 69.6^\circ$$



Your notes



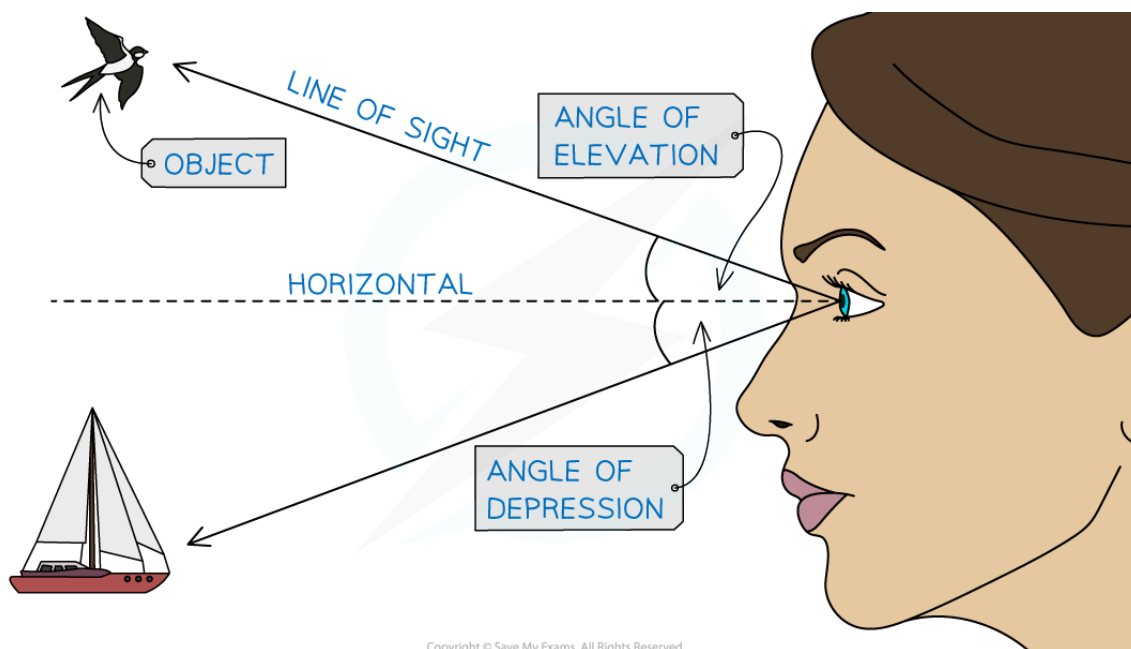
Your notes

Angles of Elevation & Depression

Elevation & Depression

What are angles of elevation and depression?

- An **angle of elevation** or **depression** is the angle measured between the **horizontal** and the **line of sight**
 - Looking **up** at an object creates an angle of **elevation**
 - Looking **down** at an object creates an angle of **depression**
- Right-angled trigonometry** can be used to find
 - an **angle** of elevation or depression
 - or a missing **distance**
- The **tan ratio** is often used in real-life scenarios
 - You may **know the height** of an object and want to **find the distance** you are from it
 - You may **know the distance** you are from an object and want to **find its height**



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Examiner Tips and Tricks

It may be useful to draw more than one diagram if the triangles that you are interested in overlap one another.



Worked Example

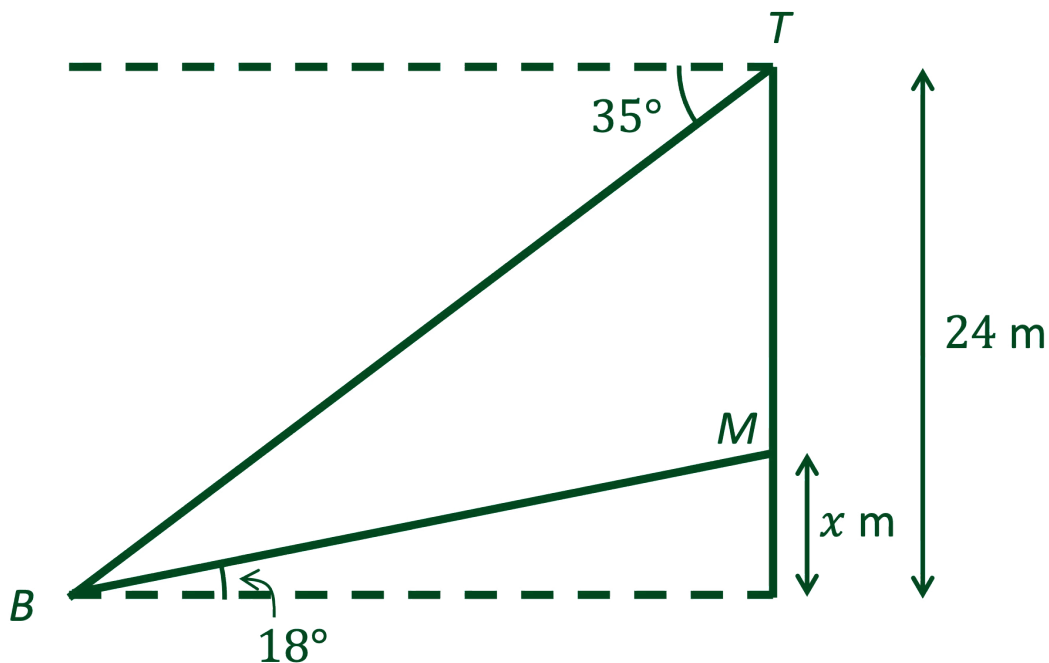
A cliff is perpendicular to the sea and the top of the cliff, T , stands 24 metres above the level of the sea.

The angle of depression from the top of the cliff to a boat at sea is 35° .

At a point X metres vertically up from the foot of the cliff is a flag marker, M .

The angle of elevation from the boat, B , to the flag marker is 18° .

(a) Draw a diagram of the situation. Label all the angles and distances given above.



(b) Find the distance from the boat to the foot of the cliff.



Your notes



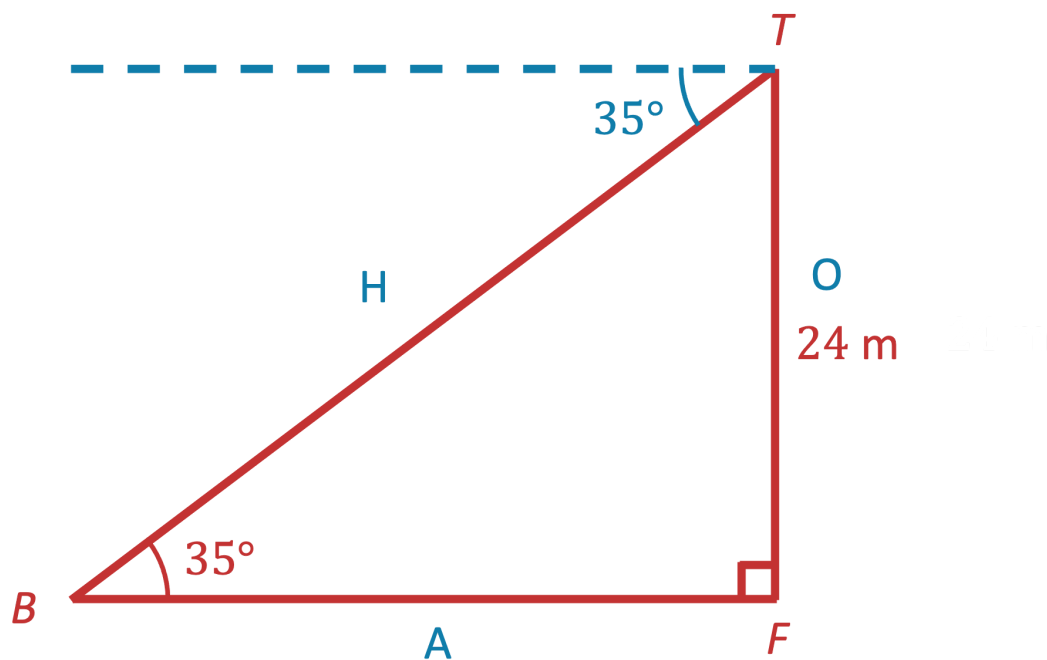
Your notes

Consider triangle TBF where F is the foot of the cliff

Angle $TBF = 35^\circ$ because of alternate angles

Use SOHCAHTOA to find the missing distance

We know the opposite (TF) and we want to find the adjacent (BF), so use $\tan \theta = \frac{O}{A}$



$$\tan 35 = \frac{24}{BF}$$

$$BF = \frac{24}{\tan 35}$$

$$BF = 34.27555...$$

$$BF = 34.3 \text{ m (3 s.f.)}$$

(c) Find the value of X .

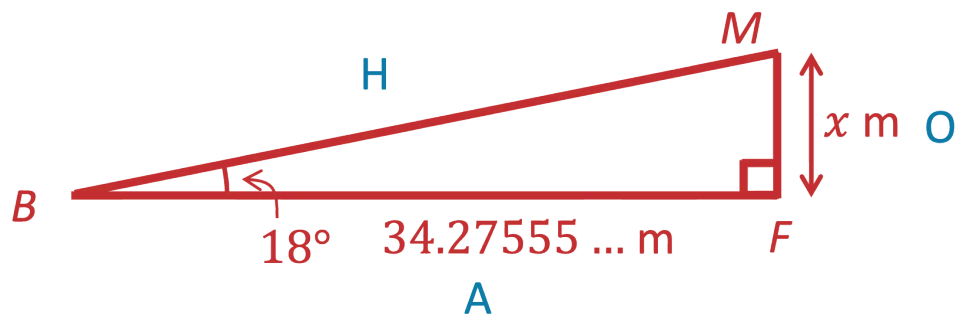
Consider triangle MBF



Your notes

Use SOHCAHTOA to find the missing distance

We know the adjacent (BF) and we want to find opposite (MF), so use $\tan \theta = \frac{O}{A}$



$$\tan 18 = \frac{x}{34.27555...}$$

$$34.27555... \tan 18 = x$$

$$x = 11.1368...$$

$$\mathbf{x = 11.1 \text{ m (3 s.f.)}}$$



Your notes

Exact Trig Values

Exact Trig Values

What are exact values in trigonometry?

- For **certain angles** the values of $\sin \theta$, $\cos \theta$ and $\tan \theta$ can be written **exactly**
 - This means using fractions and surds
- You are **expected to know** the exact values of $\sin$, $\cos$ and $\tan$ for
 - 0° , 30° , 45° , 60° , 90° , 180° and their multiples

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined

How can I remember these exact trig values?

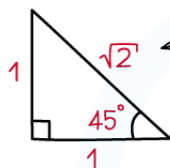
- Look at **patterns** in the table
 - Note the values of $\sin \theta$ from 0° to 90° match $\cos \theta$ in reverse, from 90° to 0°
 - Some people remember $\sin \theta$ using the trick $\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$ which simplifies to $0, \frac{1}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, 1$
- Two special **right-angled triangles** below can help you to find the exact values for 30° , 45° and 60°



Your notes

- Remember that by rationalising the denominator, $\frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

ISOSCELES RIGHT-ANGLED TRIANGLE
WITH EQUAL SIDES LENGTH 1



PYTHAGORAS' THEOREM
HYPOTENUSE = $\sqrt{2}$

$$\sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$\cos 45^\circ = \frac{1}{\sqrt{2}}$$

$$\tan 45^\circ = 1$$

CALCULATOR ANSWER IS $\frac{\sqrt{2}}{2}$,
DO YOU KNOW WHY?

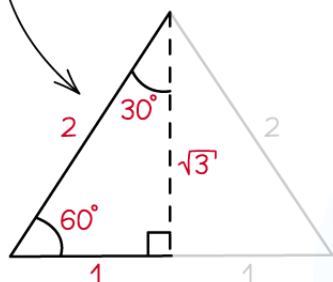
ANSWER: DENOMINATOR RATIONALISED

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Your notes

EQUILATERAL TRIANGLE
OF SIDE LENGTH 2



BY SYMMETRY, ANGLES ARE 30° , 60°
AND SIDE LENGTHS 1, 2
BY PYTHAGORAS', THIRD LENGTH IS $\sqrt{3}$

$$\sin 30^\circ = \frac{1}{2}$$

$$\sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$\cos 60^\circ = \frac{1}{2}$$

$$\tan 30^\circ = \frac{1}{\sqrt{3}}$$

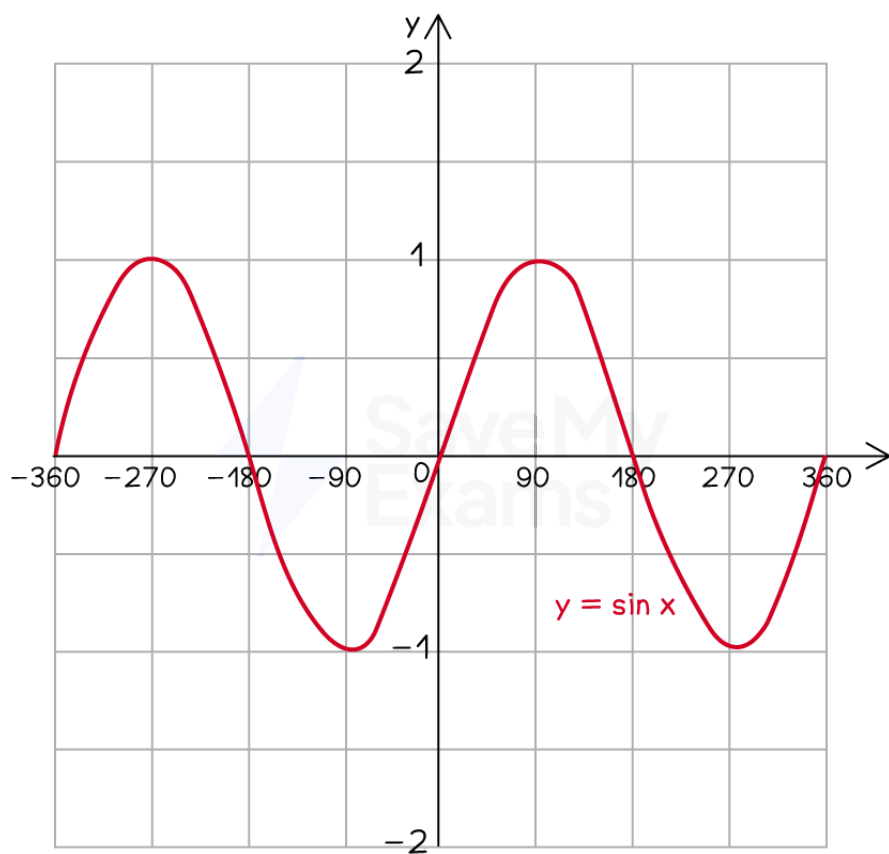
$$\tan 60^\circ = \sqrt{3}$$

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- The **trig graphs** can help you to remember the exact values for 0° and multiples of 90°



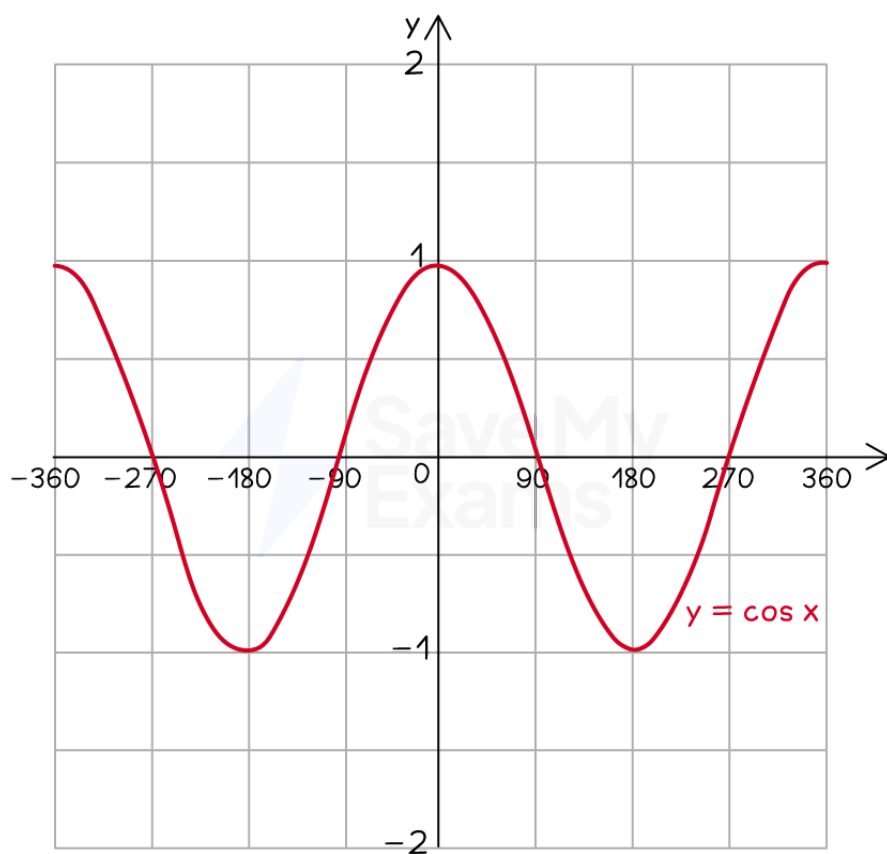
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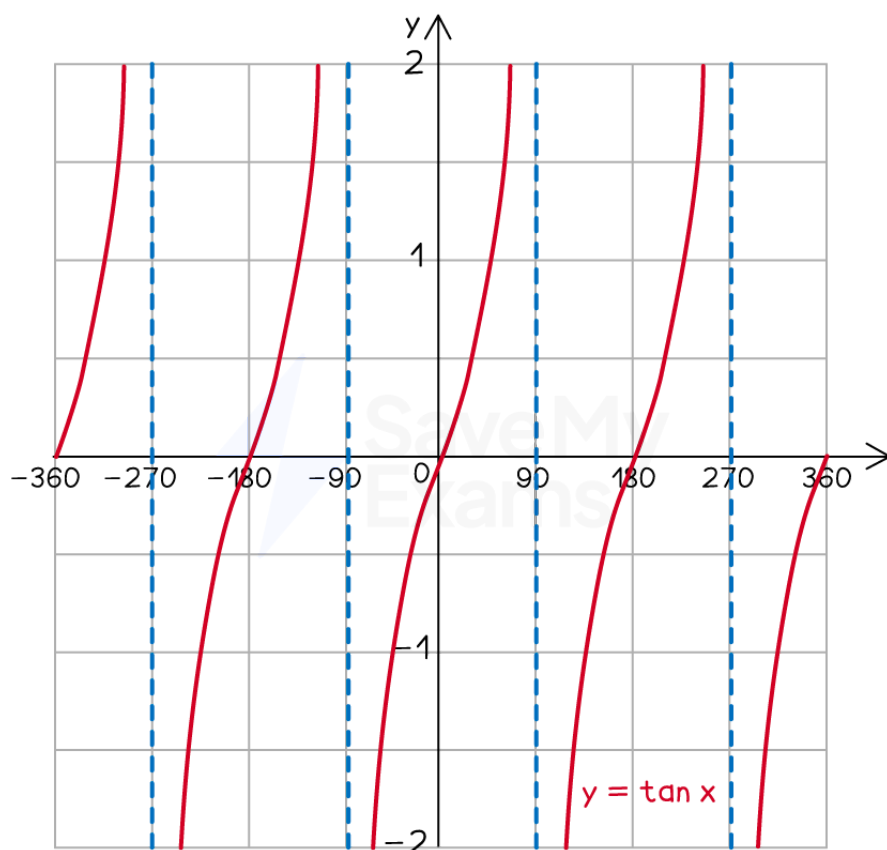
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Your notes



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How do I use exact trig values?

- You may come across trig questions in a **non-calculator** question
- In trig calculations, **substitute** in the **exact trig values** and solve as usual

- E.g. Solve the equation $\cos 45 = \frac{x}{12}$

- Replace $\cos 45$ with $\frac{\sqrt{2}}{2}$ to give $\frac{\sqrt{2}}{2} = \frac{x}{12}$

- Then you can solve for x

- On trig graphs, you may be expected to find a coordinate

- E.g. The coordinates $(30, k)$ lie on the graph $y = \tan x$, find k

- k will be equal to $\tan 30$
- The exact value of $\tan 30$ is $\frac{\sqrt{3}}{3}$
- Therefore $k = \frac{\sqrt{3}}{3}$



Examiner Tips and Tricks

- Writing these out (or sketching the triangles/graphs) on your paper at the beginning of the exam means that you can use them as many times as you need to during the exam!



Worked Example

Find the value of X in the diagram below.

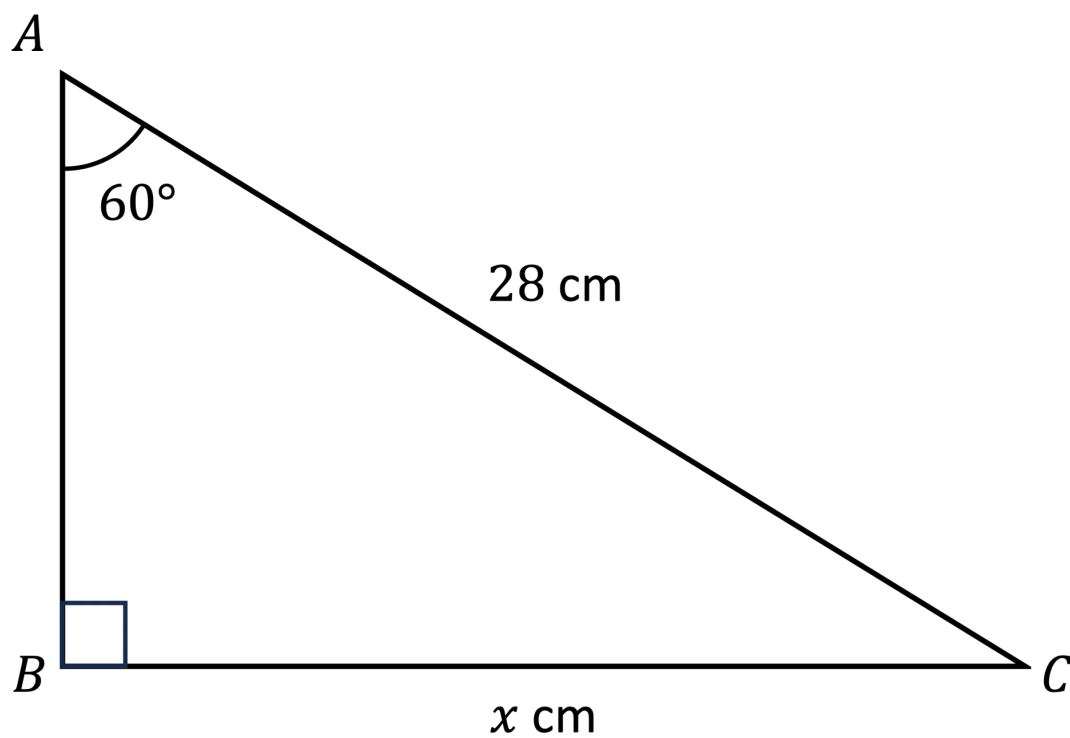
Give your answer as an exact value.



Your notes



Your notes



Triangle ABC is a right-angled triangle, so use SOHCAHTOA

We know the hypotenuse (AC) and we want to calculate the opposite (BC), so use $\sin \theta = \frac{O}{H}$

$$\sin 60 = \frac{x}{28}$$

Remember that $\sin 60 = \frac{\sqrt{3}}{2}$

So,



Your notes

$$\frac{\sqrt{3}}{2} = \frac{x}{28}$$

$$28 \times \frac{\sqrt{3}}{2} = x$$

$$14 \times \sqrt{3} = x$$

Leave in exact (surd) form

$$x = 14\sqrt{3} \text{ cm}$$